Rishi More

Sunnyvale, CA | rishi.more102@gmail.com | +1(650)220-9450 | linkedin.com/in/rishimore102 | github.com/rishi-more-2003 | rishi-more-2003.github.io

Summary

ML systems engineer and Johns Hopkins CS graduate focused on LLM inference, CUDA/Triton kernels, quantization, and adaptive serving. **ICML 2026 co-first author** with vLLM integration, Nsight profiling, and correctness-gated benchmarks spanning weight bandwidth and KV-cache memory

Education

Johns Hopkins University — **August 2024 – May 2026**
M.S.E. in Computer Science — *GPA: 3.9/4.0*

University of Mumbai — **July 2020 – July 2024**
B.Tech. in Computer Science, Honors in AI/ML — *CGPA: 9.85/10*

Technical Skills

Languages: Python, C++, CUDA C++, C, SQL, Bash
ML & Inference: PyTorch, Triton, vLLM, Hugging Face Transformers, TensorFlow, FastAPI, LLM serving, model inference
GPU Performance: CUDA, cuBLAS, Nsight Compute/Systems, roofline analysis, kernel fusion, INT4/INT8 quantization, KV cache
Infrastructure: Linux, Docker, GitHub Actions CI/CD, AWS, pytest, mypy, Prometheus, Grafana, conformal risk control

Experience

InterSystems — **July 2026 – September 2026**
Software Development Engineer, AI Experience (Core Development Program) — *Boston, Massachusetts*

- Fine-tuned **Qwen2.5-32B** with reinforcement learning from human feedback (**RLHF**) on clinician preference data, reaching **93% accuracy** on a clinician-workflow golden dataset.
- Shipped an **AI-generated patient-summary column** in TrakCare, InterSystems' electronic health record (EHR), consolidating patient histories to simplify clinician chart review and reduce navigation.

Apple ML Research + Johns Hopkins University Data Science and AI Institute — **May 2025 – June 2026**
Graduate Research Assistant, LLM Inference Optimization — *Baltimore, Maryland*

- Co-first authored *Conformal Thinking* (**ICML 2026 main track**); designed **dual-threshold early stopping** for reasoning LLMs that exits on confident answers or stalled reasoning, cutting test-time reasoning tokens **12% at matched accuracy**.
- Built a vLLM calibration and evaluation pipeline over **4 reasoning models, 4 benchmarks, and 2.7K+ problems**, logging intermediate answers, confidence trajectories, and token counts to compare stopping policies.
- Implemented **UCB-calibrated threshold selection** that picks the lowest-token policy meeting a user-set risk budget; stress-tested false-positive and premature-stop risk under dataset and response-length shift.

Mastek — **November 2022 – July 2023**
Machine Learning Engineer - Undergraduate Co-op — *Mumbai, India*

- Deployed an on-premises edge voice-authentication system with **sub-100 ms p99 latency** and **96.6% accuracy**; a lightweight Siamese mel-spectrogram model enrolled new speakers without retraining.
- Built a TensorFlow data pipeline over **1M+ voice samples**; spectral-gating denoising, augmentation, and robust feature extraction improved authentication reliability **20%** in high-noise conditions.

Projects

Decode Roofline: Kernel Profiling & Fused INT4 Dequant-GEMV | *CUDA C++, Nsight, PyTorch* | GitHub
- Profiled Qwen2.5-1.5B batch-1 decode with Nsight on an RTX 4070 Laptop GPU; attributed **81% of GPU kernel time** to cuBLAS weight GEMVs running at **247 GB/s, 95% of peak DRAM bandwidth**.
- Wrote a fused CUDA INT4 group-dequantization + GEMV kernel (FP32 accumulation) that moves **8.7x fewer bytes** than a two-op dequant path and sustains **223 GB/s (86% of peak)**; **2.38x faster** than FP16 cuBLAS GEMM at batch 1.
- Correctness-gated every timing and swept batch and shape regimes, locating the **batch-2 crossover** where weight reuse makes FP16 GEMM the better primitive.

KV Roofline: INT4 KV-Cache Quantization & Fused Decode Attention | *Triton, PyTorch, CUDA* | GitHub
- Implemented KIVI-style **per-channel K / per-token V** INT4 KV-cache quantization with an FP16 residual window; cut KV memory **3.47x** (28.7 to 8.3 KB/token at 4K), raising max context on 8 GB from **154K to 580K tokens**.
- Wrote a fused Triton dequant-attention decode kernel with token-axis splits (flash-decoding) and online softmax, **up to 27x faster** than PyTorch SDPA at 128K context (1.2 to 33.5 tok/s) on occupancy-bound batch-1 GQA decode.
- Gated kernels at **0.999+ cosine** vs. a dequantized oracle; INT4 kept **100% passkey retrieval** to 32K at **+1.0% perplexity**, while an ablation showed per-token K collapsing to 0.08 cosine under outlier channels.

Conformal Serve: Risk-Controlled Adaptive LLM Serving | *vLLM, FastAPI, Docker Compose, Prometheus, Grafana* | GitHub
- Built an OpenAI-compatible inference server on **vLLM AsyncLLMEngine** that stops reasoning under a client-set risk level, reusing KV cache in one streaming request; **78 tests** run GPU-free in CI on a mock engine.
- Calibrated thresholds offline with distribution-free risk control into content-addressed artifacts; cut reasoning tokens **64.7% at 100% accuracy** (risk 0.10) and **71.4%** (risk 0.30) on Qwen2.5-1.5B-Instruct.
- Shipped Docker Compose with Prometheus latency and token metrics, Grafana dashboards, Locust load tests, and a one-command benchmark reporting **95% Clopper-Pearson** error bounds; **100% policy branch coverage**.